MODULAR KOSZUL DUALITY FOR CHIRAL ALGEBRAS ON ALGEBRAIC CURVES: A PROGRAMME

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Abstract. Modular Koszul duality for chiral algebras is E_1–E_1 operadic Koszul duality transported into the homotopical modular chiral realm on algebraic curves. This first section of the programme summary establishes the bar construction on an algebraic curve: the logarithmic propagator $\eta_{12} = d\log(z_1 - z_2)$, the Arnold relation on three points, and the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct, together with the proof that $d^2 = 0$ at genus zero through cancellation of Arnold with the Borcherds identity. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ appears as the Σ_n-coinvariant shadow obtained by the lossy averaging map; the ordered bar is the primitive object and all five theorems of the programme extract Σ_n-invariant content from it.

1. The bar construction on algebraic curves

The bar construction of an associative algebra is classical: given an augmented algebra A over a field, form the tensor coalgebra $T^c(s^{-1}\bar{A}) = \bigoplus_{n\geq 1}(s^{-1}\bar{A})^{\otimes n}$ on desuspended generators, equip it with the differential that encodes the multiplication, and observe that $d^2 = 0$ is equivalent to associativity. Koszul duality, homological algebra, and deformation theory all descend from this single construction.

A chiral algebra on a smooth algebraic curve X encodes operator product singularities: the product of two sections $a(z)\, b(w)$ diverges as $z \to w$, and the divergence carries algebraic data. The problem is to build the bar construction in this setting. The point is replaced by a curve; the multiplication map is replaced by a chiral bracket with poles along the diagonal; the tensor coalgebra must account for the geometry of colliding points on X.

1.1. **The propagator.** On the configuration space $C_2(X) = X \times X \setminus \Delta$ of two distinct points on a smooth algebraic curve X over $\mathbf{C}$, define

$$\eta_{12} \;=\; d\log(z_1 - z_2) \;=\; \frac{dz_1 - dz_2}{z_1 - z_2}\,. \tag{1.1}$$

This 1-form has a simple pole along the diagonal $\Delta \subset X \times X$ with residue 1. The residue is intrinsic: it depends on no coordinate, no metric, no level.

Three properties single out η_{12}. A double pole $dz/(z-w)^2$ transforms under coordinate change and has no coordinate-independent residue. A regular 1-form has residue zero and extracts nothing from the collision. The logarithmic derivative is the unique 1-form on $C_2(X)$ with a first-order pole along Δ and conformally invariant residue.

On $C_3(X)$, the three pullbacks $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy

$$\eta_{12}\wedge\eta_{23} \;+\; \eta_{23}\wedge\eta_{31} \;+\; \eta_{31}\wedge\eta_{12} \;=\; 0\,. \tag{1.2}$$

This is the *Arnold relation*. It restates the partial-fractions identity

$$\frac{1}{(z_1-z_2)(z_2-z_3)} + \frac{1}{(z_2-z_3)(z_3-z_1)} + \frac{1}{(z_3-z_1)(z_1-z_2)} = 0\,,$$

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which holds because the left side, viewed as a rational function of z_1 with z_2, z_3 fixed, has simple poles at z_2 and z_3 whose residues sum to zero. Arnold [1] proved that (1.2), replicated across all triples $\{i, j, k\} \subset \{1, \dots, n\}$, generates the complete ideal of relations in $H^*(\mathrm{Conf}_n(\mathbf{C}), \mathbf{C})$.

Under a coordinate change $z \mapsto w(z)$, the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d \log \frac{w(z_i)-w(z_j)}{z_i - z_j}$. Near the diagonal the correction becomes $d \log w'(z_i)$: the standard cocycle for $\mathrm{Diff}(X)$. The propagator is a BRST ghost for diffeomorphisms; its failure to be coordinate-invariant is the failure that the Virasoro algebra measures.

1.2. **Chiral algebras.** A *chiral algebra* $\mathcal{A}$ on X is a $\mathcal{D}_X$-module equipped with a chiral bracket

$$\mu^{\mathrm{ch}} \; : \; j_* j^* (\mathcal{A} \boxtimes \mathcal{A}) \;\longrightarrow\; \Delta_! (\mathcal{A}) \,,$$

where $j \colon X \times X \setminus \Delta \hookrightarrow X \times X$ is the open complement and $\Delta_! \mathcal{A}$ is the !-pushforward along the diagonal embedding [2]. The source consists of sections of $\mathcal{A} \boxtimes \mathcal{A}$ with poles of arbitrary order along Δ; the chiral bracket extracts the singular part and projects it onto the diagonal. In local coordinates, the chiral bracket encodes the operator product expansion:

$$a(z)\, b(w) \;\sim\; \sum_{n \geq 0} \frac{a_{(n)} b(w)}{(z-w)^{n+1}} \,.$$

The n-th products $a_{(n)} b \in \mathcal{A}$ are the Fourier modes, extracted by $a_{(n)} b = \mathrm{Res}_{z=w} [a(z)\, b(w)\, (z-w)^n]$. The zeroth product $a_{(0)} b$ is the Lie bracket (first-order pole); $a_{(1)} b$ carries the symmetric/metric data (second-order pole). Higher products $a_{(n)} b$ for $n \geq 2$ are determined by the $\mathcal{D}_X$-module structure.

The Borcherds identity

$$\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c \;=\; \sum_{j \geq 0} (-1)^j \binom{n}{j} \Big(a_{(m+n-j)} (b_{(k+j)} c) - (-1)^n b_{(n+k-j)} (a_{(m+j)} c) \Big) \tag{1.3}$$

is the complete axiom system. At $n = 0$ it reduces to the Jacobi identity for the Lie bracket $a_{(0)}$; at $n = 1$, $m = k = 0$, to the derivation property of $a_{(0)}$ with respect to $a_{(1)} b$.

1.3. **The bar complex.** Place sections $a_1, \dots, a_n \in \mathcal{A}$ at distinct points $z_1, \dots, z_n \in X$. The Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(X)$ resolves $\mathrm{Conf}_n(X)$ by iterated real oriented blowup along all diagonal strata [6]. Points of $\overline{\mathrm{FM}}_n(X) \setminus \mathrm{Conf}_n(X)$ record the limiting direction and relative scale of approach as points collide. The boundary is a manifold with corners; codimension-one faces D_S are indexed by subsets $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$, recording which points have collided. Each face factors:

$$D_S \;\simeq\; \overline{\mathrm{FM}}_{|S|} \times \overline{\mathrm{FM}}_{n-|S|+1} \,,$$

cluster times residual.

The *ordered bar complex* is

$$\overline{B}{}^{\mathrm{ord}}_X (\mathcal{A}) \;=\; \bigoplus_{n \geq 1} \big(s^{-1} \bar{\mathcal{A}} \big)^{\otimes n} \;\otimes\; \Gamma\big(\overline{\mathrm{FM}}_n(X),\; \Omega^{n-1}_{\log} \big),$$

with no Σ_n-quotient. Here $\bar{\mathcal{A}} = \ker(\varepsilon \colon \mathcal{A} \to \omega_X)$ is the augmentation ideal, s^{-1} denotes desuspension ($|s^{-1} a| = |a| - 1$), and the cohomological convention $|d| = +1$ is used throughout.

The bar differential decomposes:

$$d_{\overline{B}} \;=\; d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}} \,.$$

The internal differential d_{int} acts on each tensor factor; the de Rham differential d_{dR} acts on logarithmic forms on $\overline{\mathrm{FM}}_n(X)$; and the residue differential d_{res} extracts OPE data at collision divisors. For a codimension-one face $D_{\{i,j\}}$ corresponding to the collision $z_i \to z_j$, the residue differential extracts the

p-th Fourier mode $a_{(p)}b$ of the OPE, where p is the pole order read from the form-theoretic coefficient of η_{ij}. The bar differential does not merely detect singularities; it extracts them, mode by mode, through the logarithmic kernel (1.1).

1.4. **Why $d^2 = 0$.** Expand $(d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}})^2$ into nine terms. Three vanish individually: $d_{\mathrm{int}}^2 = 0$ (the underlying cochain complex), $d_{\mathrm{dR}}^2 = 0$ (de Rham), and $d_{\mathrm{int}}\, d_{\mathrm{dR}} + d_{\mathrm{dR}}\, d_{\mathrm{int}} = 0$ (Leibniz). The remaining terms cancel pairwise, governed by index overlap:

(a) *Disjoint supports.* When $\{i,j\} \cap \{k,l\} = \emptyset$, the faces $D_{\{i,j\}}$ and $D_{\{k,l\}}$ are transverse in $\overline{\mathrm{FM}}_n(X)$. The residue operators commute: $d_{\mathrm{res}}^{(ij)}\, d_{\mathrm{res}}^{(kl)} + d_{\mathrm{res}}^{(kl)}\, d_{\mathrm{res}}^{(ij)} = 0$.

(b) *Shared index.* When $\{i,j\} \cap \{k,l\} = \{j = k\}$, the triple collision $z_i = z_j = z_l$ sits in a codimension-two stratum. The form-theoretic contribution involves $\eta_{ij} \wedge \eta_{jl}$. The Arnold relation (1.2) identifies this with a cyclic sum over the three ordered approaches to the triple-collision point. On the algebraic side, the three terms produce $(a_{i(m)}a_j)_{(n)}a_l$, $(a_{j(m)}a_l)_{(n)}a_i$, $(a_{l(m)}a_i)_{(n)}a_j$ with appropriate pole orders, whose sum vanishes by the Borcherds identity (1.3).

The Arnold relation kills the form; the Borcherds identity kills the algebra. Both are needed, and both are consumed.

(c) *Same pair.* $d_{\mathrm{res}}^{(ij)}\, d_{\mathrm{res}}^{(ij)} = 0$ by the sign rule on desuspensions: extracting a residue twice at the same diagonal vanishes for degree reasons.

The result: $d_{\overline{B}}^2 = 0$ if and only if the n-th products satisfy (1.3). The bar construction accepts exactly chiral algebras.

1.5. **Two coproducts.** The ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ carries the *deconcatenation coproduct*:

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] \;=\; \sum_{i=0}^{n}[s^{-1}a_1|\cdots|s^{-1}a_i] \;\otimes\; [s^{-1}a_{i+1}|\cdots|s^{-1}a_n]\,,$$

splitting an ordered sequence at every consecutive position: $n+1$ terms. This is coassociative but not cocommutative. It makes $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ a cofree tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$: the E_1-coalgebra reflecting the linear ordering of points.

The Σ_n-coinvariant quotient

$$\overline{B}_X^{\Sigma}(\mathcal{A}) \;=\; \bigoplus_{n\geq 1}(s^{-1}\bar{\mathcal{A}})^{\otimes n}_{\Sigma_n} \;\otimes\; \Gamma\big(\overline{\mathrm{FM}}_n(X),\; \Omega^{n-1}_{\log}\big)$$

carries the *coshuffle coproduct*, summing over all 2^n bipartitions of an unordered set: coassociative and cocommutative. This is the factorization coalgebra of Beilinson–Drinfeld on $\mathrm{Ran}(X)$ [2], the E_∞-coalgebra.

The ordered bar is the level-2 chain object before averaging. The symmetric bar is a quotient that loses data: the non-cocommutative structure of the ordered bar carries the spectral R-matrix, the Knizhnik–Zamolodchikov associator, and the full quantum group deformation, none of which survive the passage to Σ_n-coinvariants.

Remark 1.1 (Six bar complexes). Six bar-type objects coexist. The Francis–Gaitsgory bar [5] $B^{\mathrm{Lie}}(\mathcal{A})$ is the cofree coLie coalgebra $\mathrm{Lie}^c(s^{-1}\bar{\mathcal{A}})$, capturing zeroth-pole data; it embeds into $\overline{B}^{\Sigma}$ via the inclusion $\mathrm{Lie}^c \hookrightarrow \mathrm{Sym}^c$ (a quasi-isomorphism in characteristic zero, but a categorical distinction). The genus-graded bar $B^{(g)}(\mathcal{A})$ extends $\overline{B}^{\Sigma}$ to families of curves over $\overline{\mathcal{M}}_g$, where the differential acquires curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ (see §1.8). The modular bar $B^{\mathrm{mod}}(\mathcal{A})$ restores $D^2 = 0$ by completing over the Feynman transform of the commutative modular operad. The E_1-modular bar $B^{E_1\text{-}\mathrm{mod}}(\mathcal{A})$, an algebra over the

Feynman transform of the associative modular operad, is the primitive among these; the modular bar is its Σ_n-coinvariant image. No two of these six objects are isomorphic in general.

1.6. **Three operations on the bar.** The bar construction is a categorical logarithm: $B(\mathcal{A} \otimes \mathcal{A}') \simeq B(\mathcal{A}) \oplus B(\mathcal{A}')$. The logarithmic form $\eta_{ij} = d\log(z_i - z_j)$ is the integral kernel; the Arnold relation is the cocycle condition for log.

Three distinct functors act on $B(\mathcal{A})$ and produce three distinct outputs:

$$\begin{aligned} \Omega(B(\mathcal{A})) &\simeq \mathcal{A} && \text{(cobar inversion: recovers the original),} \\ \mathbb{D}_{\mathrm{Ran}}(B(\mathcal{A})) &\simeq \mathcal{A}^!_\infty && \text{(Verdier duality: the homotopy Koszul dual),} \\ (H^*(B(\mathcal{A})))^\vee &= \mathcal{A}^! && \text{(linear duality of bar cohomology: the strict dual).} \end{aligned} \tag{1.4}$$

Cobar inversion recovers the original algebra $\mathcal{A}$ itself; it is a round-trip, not a duality. Verdier duality on $\mathrm{Ran}(X)$ converts the bar coalgebra into a factorization algebra $\mathcal{A}^!_\infty$, the homotopy Koszul dual retaining full chain-level data. That its cohomology recovers the strict Koszul dual $\mathcal{A}^!$ on the Koszul locus is the content of the Verdier intertwining theorem, not a tautology. Confusing any two of (1.4) is a category error.

1.7. **The Heisenberg algebra at bar degree two.** The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single field α of conformal weight 1 with OPE

$$\alpha(z)\,\alpha(w) \;\sim\; \frac{k}{(z-w)^2}\,.$$

Only the second-order pole is present: $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$ and $\alpha_{(n)}\alpha = 0$ for $n \neq 1$. No first-order pole: $\alpha_{(0)}\alpha = 0$, no Lie bracket. All triple products vanish since $\mathbf{1}_{(n)} = 0$ for $n \geq 0$.

The bar complex at tensor degree 2 gives

$$d_{\mathrm{res}}(s^{-1}\alpha \otimes s^{-1}\alpha \otimes \eta_{12}) \;=\; \alpha_{(1)}\alpha \;=\; k \cdot \mathbf{1}\,.$$

The level k is visible already at genus 0. At tensor degree $n \geq 3$: every residue at a collision stratum $D_{\{i,j\}}$ produces $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$ on the pair (i,j); since $\mathbf{1}_{(p)}\alpha = 0$ for $p \geq 0$, no further contractions occur. All structure is captured at tensor degree 2: the bar complex is quadratic. The Heisenberg algebra is chirally Koszul.

The ordered bar carries the collision residue

$$r(z) \;=\; \frac{k}{z}\,.$$

The second-order OPE pole $k/(z-w)^2$ drops by one order under the logarithmic kernel $d\log(z-w)$: the bar differential absorbs one power of $(z-w)$, producing the first-order r-matrix pole. Applying the averaging map:

$$\mathrm{av}\big(r(z)\big) \;=\; k \;=\; \kappa(\mathcal{H}_k)\,.$$

For the Heisenberg algebra, the R-matrix *is* the modular characteristic: $r(z) = k/z$ is scalar, so averaging loses nothing.

The deficiency: no first-order pole, no Lie bracket, no matrix structure in the collision residue, no quantum group. Everything that makes representation theory nontrivial is absent. The simplest algebra where the R-matrix is matrix-valued is the first algebra with a first-order pole: the affine Kac–Moody algebra.

Example 1.2 (Kac–Moody collision residue)**.** The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k is generated by currents J^a, $a = 1, \ldots, \dim \mathfrak{g}$, with OPE

$$J^a(z)\, J^b(w) \;\sim\; \frac{f^{ab}{}_c\, J^c(w)}{z-w} \;+\; \frac{k\,\kappa^{ab}}{(z-w)^2}\,.$$

Both poles are present. The ordered bar carries the collision residue

$$r(z) \;=\; \frac{k\,\Omega}{z}\,, \qquad \Omega = \sum_a J^a \otimes J_a \;\in\; \mathfrak{g}\otimes\mathfrak{g}\,,$$

where Ω is the Casimir tensor and k is the level. At $k = 0$ the r-matrix vanishes: no level, no collision data. The averaging map collapses $k\,\Omega$ to a scalar:

$$\mathrm{av}(k\,\Omega/z) \;=\; \frac{k\ \dim(\mathfrak{g})}{2h^\vee} \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k)\,,$$

where $h^\vee$ is the dual Coxeter number. The full modular characteristic adds the Sugawara shift:

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})\,(k+h^\vee)}{2h^\vee}.$$

The matrix structure of Ω, erased by Σ_2-coinvariance, encodes the representation theory of $\mathfrak{g}$: it is the datum that builds the Yangian $Y(\mathfrak{g})$.

1.8. **Genus 0 and genus** ≥ 1**.** Everything above takes place at genus 0: the propagator $\eta_{12} = d\log(z_1 - z_2)$ is globally defined on $\mathbb{P}^1$, and the Arnold relation holds without correction. On a curve of genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning: the curve is compact, and any meromorphic function with a single simple zero must also have a simple pole.

The replacement is the *prime form* $E(z_1, z_2)$, a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal with no other zeros. The genus-g propagator

$$\eta_{12}^{(g)} \;=\; d\log E(z_1, z_2) + (\text{period correction})$$

is single-valued but no longer holomorphic; the period correction couples to the Hodge bundle $\mathbb{E} \to \overline{\mathcal{M}}_g$ through the period matrix. The propagator $d\log E(z_1, z_2)$ has conformal weight 1 in both variables, regardless of the conformal weight of the field being sewed: all edges of the genus-g graph sum use the standard Hodge bundle $\mathbb{E}_1 = R^0\pi_*\omega_\pi$.

The Arnold relation acquires a defect, and the fibrewise bar differential satisfies

$$d_{\mathrm{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_g\,,$$

where ω_g is the Arakelov form on the fibre of $\pi : \mathcal{C}_g \to \overline{\mathcal{M}}_g$ and $\kappa(\mathcal{A}) \in \mathbf{C}$ is the *modular characteristic*, determined entirely by the leading OPE singularity. Explicit values:

$$\kappa(\mathcal{H}_k) = k\,, \qquad \kappa(\widehat{\mathfrak{g}}_k) = \frac{\dim(\mathfrak{g})\cdot(k+h^\vee)}{2h^\vee}\,, \qquad \kappa(\mathrm{Vir}_c) = \frac{c}{2}\,.$$

Period integrals restore nilpotence: the Gauss–Manin connection absorbs the fibrewise curvature, and the total differential satisfies $D_g^2 = 0$.

Genus 0 is uncurved algebra. Genus ≥ 1 is curved geometry. The entire genus tower is controlled by a single scalar: the curvature of the bar complex on a family of curves.

Remark 1.3 (Recovering the classical bar). When $X = \operatorname{Spec}\mathbf{C}$ is a point, configuration spaces collapse, logarithmic forms vanish, and the chiral bar complex recovers the classical bar construction of an associative algebra A. This recovery requires specifying the retraction data: vertex algebras live on the formal disk D, and the passage from D to a bare point discards the thickening (completion, growth conditions) on which the $\mathcal{D}$-module structure depends. A quadratic algebra $A = T(V)/(R)$ has quadratic dual $A^! = T(V^*)/(R^\perp)$; when A is Koszul, the bar cohomology concentrates in diagonal degrees and $H^*(B(A)) \cong (A^!)^\vee$. The operadic instances $\mathrm{Com}^! = \mathrm{Lie}$ and $\mathrm{Sym}^! = \Lambda$ recover the Chevalley–Eilenberg complex as a restricted bar construction.

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